

Q-learning Tutorial

April 6, 2026

based on: <https://www.datacamp.com/tutorial/introduction-q-learning-beginner-tutorial>

1 What is Q-Learning

```
[108]: import os
from google.colab import drive

# Mount Google Drive to save the HTML file
drive.mount('/content/drive')

# Define the input path for the current notebook and the output path for the HTML file
# This assumes your notebook is in 'My Drive/Colab Notebooks/'. Adjust if needed.
current_notebook_name = 'Q-learning Tutorial.ipynb' # Ensure this matches your notebook's name
input_notebook_path = f'/content/drive/MyDrive/Colab Notebooks/{current_notebook_name}'
output_html_path = '/content/drive/MyDrive/Q-learning_Tutorial.html'

print(f"Input notebook path: {input_notebook_path}")
print(f"Output HTML path: {output_html_path}")
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `drive.mount("/content/drive", force_remount=True)`.

Input notebook path: /content/drive/MyDrive/Colab Notebooks/Q-learning Tutorial.ipynb

Output HTML path: /content/drive/MyDrive/Q-learning_Tutorial.html

Q-learning is a **model-free, value-based, off-policy** algorithm that will find the best series of actions based on the agent's current state.

The model-based algorithms use transition and reward functions to estimate the optimal policy and create the model. In contrast, model-free algorithms learn the consequences of their actions through the experience without transition and reward function.

The value-based method trains the value function to learn which state is more valuable and take action. On the other hand, policy-based methods train the policy directly to learn which action to take in a given state

In the off-policy, the algorithm evaluates and updates a policy that differs from the policy used to take an action. Conversely, the on-policy algorithm evaluates and improves the same policy used to take an action.

2 Example w/ Pygame

```
[109]: !pip install pygame==1.5.1
!apt install python-opengl
!apt install ffmpeg
!apt install xvfb
!pip3 install pyvirtualdisplay

# Virtual display
from pyvirtualdisplay import Display

virtual_display = Display(visible=0, size=(1400, 900))
virtual_display.start()
```

Requirement already satisfied: pygame==1.5.1 in /usr/local/lib/python3.12/dist-packages (1.5.1)

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

E: Unable to locate package python-opengl

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

ffmpeg is already the newest version (7:4.4.2-0ubuntu0.22.04.1).

0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

xvfb is already the newest version (2:21.1.4-2ubuntu1.7~22.04.16).

0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.

Requirement already satisfied: pyvirtualdisplay in
/usr/local/lib/python3.12/dist-packages (3.0)

/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203:

DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).

return datetime.utcnow().replace(tzinfo=utc)

```
[109]: <pyvirtualdisplay.display.Display at 0x7f47498fe900>
```

```
[110]: !pip install gym==0.24
!pip install pygame
```

```
!pip install numpy
```

```
!pip install imageio imageio_ffmpeg
```

```
Requirement already satisfied: gym==0.24 in /usr/local/lib/python3.12/dist-packages (0.24.0)
Requirement already satisfied: numpy>=1.18.0 in /usr/local/lib/python3.12/dist-packages (from gym==0.24) (2.0.2)
Requirement already satisfied: cloudpickle>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from gym==0.24) (3.1.2)
Requirement already satisfied: gym_notices>=0.0.4 in /usr/local/lib/python3.12/dist-packages (from gym==0.24) (0.1.0)
Requirement already satisfied: pygame in /usr/local/lib/python3.12/dist-packages (2.6.1)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: imageio in /usr/local/lib/python3.12/dist-packages (2.37.3)
Requirement already satisfied: imageio_ffmpeg in /usr/local/lib/python3.12/dist-packages (0.6.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from imageio) (2.0.2)
Requirement already satisfied: pillow>=8.3.2 in /usr/local/lib/python3.12/dist-packages (from imageio) (11.3.0)
```

```
[111]: import numpy as np
import gym
import random
import imageio
from tqdm.notebook import trange
```

```
[112]: env = gym.make("FrozenLake-v1",map_name="4x4",is_slippery=False)

print("Observation Space", env.observation_space)
print("Sample observation", env.observation_space.sample()) # display a random
↳ observation
```

```
Observation Space Discrete(16)
Sample observation 3
```

```
[113]: print("Action Space Shape", env.action_space.n)
print("Action Space Sample", env.action_space.sample())
```

```
Action Space Shape 4
Action Space Sample 3
```

```
[114]: state_space = env.observation_space.n
print("There are ", state_space, " possible states")

action_space = env.action_space.n
print("There are ", action_space, " possible actions")
```

There are 16 possible states
There are 4 possible actions

```
[115]: def initialize_q_table(state_space, action_space):
    Qtable = np.zeros((state_space, action_space))
    return Qtable

Qtable_frozenlake = initialize_q_table(state_space, action_space)
```

```
[116]: def epsilon_greedy_policy(Qtable, state, epsilon):
    random_int = random.uniform(0,1)
    if random_int > epsilon:
        action = np.argmax(Qtable[state])
    else:
        action = env.action_space.sample()
    return action
```

```
[117]: def greedy_policy(Qtable, state):
    action = np.argmax(Qtable[state])
    return action
```

2.1 Model hyperparameters

There are 10,000 training and 100 evaluation episodes.

The learning rate is 0.7.

We are using “FrozenLake-v1” as an environment with 99 maximum steps per episode.

The gamma (discount rate) is 0.95.

eval_seed: evaluation seed for the environment.

The exploration epsilon probability at the start is 1.0, and the minimum probability will be 0.05.

The exponential decay rate for epsilon probability is 0.0005.

```
[118]: # Training parameters
n_training_episodes = 10000
learning_rate = 0.7

# Evaluation parameters
n_eval_episodes = 100
```

```

# Environment parameters
env_id = "FrozenLake-v1"
max_steps = 99
gamma = 0.95
eval_seed = []

# Exploration parameters
max_epsilon = 1.0
min_epsilon = 0.05
decay_rate = 0.0005

```

2.2 Training tldr;

Create a loop for training episodes.

We will first reduce epsilon. As we need less and less exploration and more exploitation with every episode.

Reset the environment.

Create a nested loop for the maximum steps.

Choose the action using the epsilon greedy policy.

Take action (A_t) and observe the expected reward (R_{t+1}) and state (S_{t+1}).

Take the action (a) and observe the outcome state (s') and reward (r).

Update the Q-function using the formula.

If done= True, finish the episode and break the loop.

Finally, change the current state to a new state.

After completing all of the training episodes, the function will return the updated Q-Table.

```

[119]: def train(n_training_episodes, min_epsilon, max_epsilon, decay_rate, env, Q
    ↪max_steps, Qtable):
    for episode in trange(n_training_episodes):

        epsilon = min_epsilon + (max_epsilon - min_epsilon)*np.
    ↪exp(-decay_rate*episode)
        # Reset the environment
        state = env.reset()
        step = 0
        done = False

        # repeat
        for step in range(max_steps):

            action = epsilon_greedy_policy(Qtable, state, epsilon)

```

```

    new_state, reward, done, info = env.step(action)

    Qtable[state][action] = Qtable[state][action] + learning_rate * (reward +
↪gamma * np.max(Qtable[new_state]) - Qtable[state][action])

    # If done, finish the episode
    if done:
        break

    # Our state is the new state
    state = new_state
    return Qtable

```

```

[120]: Qtable_frozenlake = train(n_training_episodes, min_epsilon, max_epsilon,
↪decay_rate, env, max_steps, Qtable_frozenlake)

```

```

0%|          | 0/10000 [00:00<?, ?it/s]

```

```

[121]: Qtable_frozenlake

```

```

[121]: array([[0.73509189, 0.77378094, 0.77378094, 0.73509189],
              [0.73509189, 0.          , 0.81450625, 0.77378094],
              [0.77378094, 0.857375   , 0.77378094, 0.81450625],
              [0.81450625, 0.          , 0.77378094, 0.77378094],
              [0.77378094, 0.81450625, 0.          , 0.73509189],
              [0.          , 0.          , 0.          , 0.          ],
              [0.          , 0.9025    , 0.          , 0.81450625],
              [0.          , 0.          , 0.          , 0.          ],
              [0.81450625, 0.          , 0.857375   , 0.77378094],
              [0.81450625, 0.9025    , 0.9025    , 0.          ],
              [0.857375   , 0.95       , 0.          , 0.857375   ],
              [0.          , 0.          , 0.          , 0.          ],
              [0.          , 0.          , 0.          , 0.          ],
              [0.          , 0.9025    , 0.95       , 0.857375   ],
              [0.9025    , 0.95       , 1.          , 0.9025    ],
              [0.          , 0.          , 0.          , 0.          ]])

```

2.3 Visualizing result as GIF

```

[122]: def record_video(env, Qtable, out_directory, fps=1):
        images = []
        done = False
        state = env.reset(seed=random.randint(0,500))
        img = env.render(mode='rgb_array')
        images.append(img)

```

```

while not done:
    # Take the action (index) that have the maximum expected future reward
    ↪given that state
    action = np.argmax(Qtable[state][:])
    state, reward, done, info = env.step(action) # We directly put next_state =
    ↪state for recording logic
    img = env.render(mode='rgb_array')
    images.append(img)
    imageio.mimsave(out_directory, [np.array(img) for i, img in
    ↪enumerate(images)], fps=fps)

```

```

[123]: video_path="/content/replay.gif"
video_fps=1
record_video(env, Qtable_frozenlake, video_path, video_fps)

from IPython.display import Image
Image('./replay.gif')

```

/usr/local/lib/python3.12/dist-packages/imageio/plugins/pillow.py:410:
DeprecationWarning: The keyword `fps` is no longer supported. Use `duration` (in
ms) instead, e.g. `fps=50` == `duration=20` (1000 * 1/50).
warnings.warn(

```

[123]: <IPython.core.display.Image object>

```

```

[124]: # Use nbconvert to convert the notebook to HTML
# The --output option specifies the output file path
# By default, nbconvert includes cell outputs when converting to html
[!]jupyter nbconvert "{input_notebook_path}" --to html --output
    ↪"{output_html_path}"

print(f"Notebook converted to HTML and saved to: {output_html_path}")
print("You can now find 'Q-learning_Tutorial.html' in your Google Drive.")

```

[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab
Notebooks/Q-learning Tutorial.ipynb to html
[NbConvertApp] Writing 316756 bytes to
/content/drive/MyDrive/Q-learning_Tutorial.html
Notebook converted to HTML and saved to:
/content/drive/MyDrive/Q-learning_Tutorial.html
You can now find 'Q-learning_Tutorial.html' in your Google Drive.

```

[ ]: # Install necessary packages for PDF conversion (TeX Live and Pandoc)
# This might take a few minutes.
[!]apt-get update
[!]apt-get install texlive-xetex texlive-fonts-recommended texlive-plain-generic
[!]apt-get install pandoc

```

```
# Also ensure jupyter is installed, though it usually is in Colab
```

```
!pip install jupyter
```

```
0% [Working]          Get:1 http://security.ubuntu.com/ubuntu jammy-  
security InRelease [129 kB]  
0% [Connecting to archive.ubuntu.com (91.189.91.81)] [1 InRelease 0 B/129 kB 0%  
Get:2 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease  
[3,632 B]  
Hit:3 https://cli.github.com/packages stable InRelease  
Hit:4 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease  
Hit:5 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease  
Hit:6 http://archive.ubuntu.com/ubuntu jammy InRelease  
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]  
Get:8 https://r2u.stat.illinois.edu/ubuntu jammy InRelease [6,555 B]  
Get:9 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]  
Get:10 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages  
[1,310 kB]  
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages  
[3,844 kB]  
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages  
[1,622 kB]  
Get:13 https://r2u.stat.illinois.edu/ubuntu jammy/main amd64 Packages [2,960 kB]  
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4,219  
kB]  
Get:15 https://r2u.stat.illinois.edu/ubuntu jammy/main all Packages [9,980 kB]  
Fetched 24.3 MB in 4s (6,657 kB/s)  
Reading package lists... Done  
W: Skipping acquire of configured file 'main/source/Sources' as repository  
'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide  
it (sources.list entry misspelt?)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono  
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java  
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12  
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0  
  libsynchronet2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13  
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet  
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils  
  teckit tex-common tex-gyre texlive-base texlive-binaries texlive-latex-base  
  texlive-latex-extra texlive-latex-recommended texlive-pictures tipa  
  xfonts-encodings xfonts-utils  
Suggested packages:  
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
```



```

libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

```

The following NEW packages will be installed:

```

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

```

0 upgraded, 53 newly installed, 0 to remove and 8 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1 [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all 0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17 [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all 20200910-1 [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1 [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64 0.35-15build2 [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64 0.19-3build2 [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.3 [60.6 kB]

Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64

1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-logging-
java all 1.2-2 [60.3 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libptexenc1
amd64 2021.20210626.59705-1ubuntu0.3 [39.1 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy/main amd64 rubygems-integration
all 1.18 [5,336 B]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby3.0 amd64
3.0.2-7ubuntu2.12 [50.1 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-rubygems
all 3.3.5-2ubuntu1.2 [228 kB]
Get:24 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:25 http://archive.ubuntu.com/ubuntu jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
Get:26 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby-net-telnet all
0.1.1-2 [12.6 kB]
Get:27 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-webrick
all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:28 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-xmlrpc all
0.3.2-1ubuntu0.1 [24.9 kB]
Get:29 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libruby3.0
amd64 3.0.2-7ubuntu2.12 [5,113 kB]
Get:30 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsynctex2
amd64 2021.20210626.59705-1ubuntu0.3 [55.8 kB]
Get:31 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libteckit0 amd64
2.5.11+ds1-1 [421 kB]
Get:32 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexlua53
amd64 2021.20210626.59705-1ubuntu0.3 [120 kB]
Get:33 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexluajit2
amd64 2021.20210626.59705-1ubuntu0.3 [267 kB]
Get:34 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzzip-0-13 amd64
0.13.72+dfsg.1-1.1 [27.0 kB]
Get:35 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:36 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64

```

1:7.7+6build2 [94.6 kB]
Get:37 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.3 [9,861 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 5s (36.2 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 118194 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...

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Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.

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Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.12_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.12) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.12_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.12) ...
Selecting previously unselected package libsyntaxtex2:amd64.
Preparing to unpack .../29-libsyntaxtex2_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libsyntaxtex2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libtexluaajit2:amd64.
Preparing to unpack
.../32-libtexluaajit2_2021.20210626.59705-1ubuntu0.3_amd64.deb ...
Unpacking libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.3) ...

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Selecting previously unselected package libzip-0-13:amd64.
 Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
 Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
 Selecting previously unselected package xfonts-encodings.
 Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
 Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
 Selecting previously unselected package xfonts-utils.
 Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
 Unpacking xfonts-utils (1:7.7+6build2) ...
 Selecting previously unselected package lmodern.
 Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
 Unpacking lmodern (2.004.5-6.1) ...
 Selecting previously unselected package preview-latex-style.
 Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
 Unpacking preview-latex-style (12.2-1ubuntu1) ...
 Selecting previously unselected package t1utils.
 Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
 Unpacking t1utils (1.41-4build2) ...
 Selecting previously unselected package teckit.
 Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
 Unpacking teckit (2.5.11+ds1-1) ...
 Selecting previously unselected package tex-gyre.
 Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
 Unpacking tex-gyre (20180621-3.1) ...
 Selecting previously unselected package texlive-binaries.
 Preparing to unpack .../41-texlive-binaries_2021.20210626.59705-1ubuntu0.3_amd64.deb ...
 Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.3) ...
 Selecting previously unselected package texlive-base.
 Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
 Unpacking texlive-base (2021.20220204-1) ...
 Selecting previously unselected package texlive-fonts-recommended.
 Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
 Unpacking texlive-fonts-recommended (2021.20220204-1) ...
 Selecting previously unselected package texlive-latex-base.
 Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
 Unpacking texlive-latex-base (2021.20220204-1) ...
 Selecting previously unselected package libfontbox-java.
 Preparing to unpack .../45-libfontbox-java_1%3a1.8.16-2_all.deb ...
 Unpacking libfontbox-java (1:1.8.16-2) ...
 Selecting previously unselected package libpdfbox-java.
 Preparing to unpack .../46-libpdfbox-java_1%3a1.8.16-2_all.deb ...
 Unpacking libpdfbox-java (1:1.8.16-2) ...
 Selecting previously unselected package texlive-latex-recommended.
 Preparing to unpack .../47-texlive-latex-recommended_2021.20220204-1_all.deb ...
 Unpacking texlive-latex-recommended (2021.20220204-1) ...
 Selecting previously unselected package texlive-pictures.
 Preparing to unpack .../48-texlive-pictures_2021.20220204-1_all.deb ...

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Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../49-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../50-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../51-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../52-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...

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Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.3) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.12) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.12) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

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/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
 This may take some time... done.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
pandoc is already the newest version (2.9.2.1-3ubuntu2).
pandoc set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
Collecting jupyter

 Downloading jupyter-1.1.1-py2.py3-none-any.whl.metadata (2.0 kB)
Requirement already satisfied: notebook in /usr/local/lib/python3.12/dist-packages (from jupyter) (6.5.7)
Requirement already satisfied: jupyter-console in /usr/local/lib/python3.12/dist-packages (from jupyter) (6.6.3)
Requirement already satisfied: nbconvert in /usr/local/lib/python3.12/dist-packages (from jupyter) (7.17.0)
Requirement already satisfied: ipykernel in /usr/local/lib/python3.12/dist-packages (from jupyter) (6.17.1)
Requirement already satisfied: ipywidgets in /usr/local/lib/python3.12/dist-

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packages (from jupyter) (7.7.1)
Collecting jupyterlab (from jupyter)
  Downloading jupyterlab-4.5.6-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: debugpy>=1.0 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (1.8.15)
Requirement already satisfied: ipython>=7.23.1 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (7.34.0)
Requirement already satisfied: jupyter-client>=6.1.12 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (7.4.9)
Requirement already satisfied: matplotlib-inline>=0.1 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (0.2.1)
Requirement already satisfied: nest-asyncio in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (1.6.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (26.0)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (5.9.5)
Requirement already satisfied: pyzmq>=17 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (26.2.1)
Requirement already satisfied: tornado>=6.1 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (6.5.1)
Requirement already satisfied: traitlets>=5.1.0 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter) (5.7.1)
Requirement already satisfied: ipython-genutils~=0.2.0 in /usr/local/lib/python3.12/dist-packages (from ipywidgets->jupyter) (0.2.0)
Requirement already satisfied: widgetsnbextension~=3.6.0 in /usr/local/lib/python3.12/dist-packages (from ipywidgets->jupyter) (3.6.10)
Requirement already satisfied: jupyterlab-widgets>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from ipywidgets->jupyter) (3.0.16)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in /usr/local/lib/python3.12/dist-packages (from jupyter-console->jupyter) (5.9.1)
Requirement already satisfied: prompt-toolkit>=3.0.30 in /usr/local/lib/python3.12/dist-packages (from jupyter-console->jupyter) (3.0.52)
Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-packages (from jupyter-console->jupyter) (2.20.0)
Collecting async-lru>=1.0.0 (from jupyterlab->jupyter)
  Downloading async_lru-2.3.0-py3-none-any.whl.metadata (7.6 kB)
Requirement already satisfied: httpx<1,>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter) (0.28.1)
Requirement already satisfied: Jinja2>=3.0.3 in /usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter) (3.1.6)
Collecting jupyter-lsp>=2.0.0 (from jupyterlab->jupyter)
  Downloading jupyter_lsp-2.3.1-py3-none-any.whl.metadata (1.8 kB)
Requirement already satisfied: jupyter-server<3,>=2.4.0 in /usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter) (2.14.0)
Collecting jupyterlab-server<3,>=2.28.0 (from jupyterlab->jupyter)
  Downloading jupyterlab_server-2.28.0-py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: notebook-shim>=0.2 in

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/usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter) (0.2.4)
Requirement already satisfied: setuptools>=41.1.0 in
/usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter) (75.2.0)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-
packages (from nbconvert->jupyter) (4.13.5)
Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.12/dist-
packages (from bleach[css]!=5.0.0->nbconvert->jupyter) (6.3.0)
Requirement already satisfied: defusedxml in /usr/local/lib/python3.12/dist-
packages (from nbconvert->jupyter) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in
/usr/local/lib/python3.12/dist-packages (from nbconvert->jupyter) (0.3.0)
Requirement already satisfied: markupsafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from nbconvert->jupyter) (3.0.3)
Requirement already satisfied: mistune<4,>=2.0.3 in
/usr/local/lib/python3.12/dist-packages (from nbconvert->jupyter) (3.2.0)
Requirement already satisfied: nbclient>=0.5.0 in
/usr/local/lib/python3.12/dist-packages (from nbconvert->jupyter) (0.10.4)
Requirement already satisfied: nbformat>=5.7 in /usr/local/lib/python3.12/dist-
packages (from nbconvert->jupyter) (5.10.4)
Requirement already satisfied: pandocfilters>=1.4.1 in
/usr/local/lib/python3.12/dist-packages (from nbconvert->jupyter) (1.5.1)
Requirement already satisfied: argon2-cffi in /usr/local/lib/python3.12/dist-
packages (from notebook->jupyter) (25.1.0)
Requirement already satisfied: Send2Trash>=1.8.0 in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter) (2.1.0)
Requirement already satisfied: terminado>=0.8.3 in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter) (0.18.1)
Requirement already satisfied: prometheus-client in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter) (0.24.1)
Requirement already satisfied: nbclassic>=0.4.7 in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter) (1.3.3)
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-
packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert->jupyter) (0.5.1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from
bleach[css]!=5.0.0->nbconvert->jupyter) (1.4.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages
(from httpx<1,>=0.25.0->jupyterlab->jupyter) (4.13.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.25.0->jupyterlab->jupyter) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.25.0->jupyterlab->jupyter) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages
(from httpx<1,>=0.25.0->jupyterlab->jupyter) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-
packages (from httpcore==1.*->httpx<1,>=0.25.0->jupyterlab->jupyter) (0.16.0)
Collecting jedi>=0.16 (from ipython>=7.23.1->ipykernel->jupyter)
  Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)

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Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-packages (from ipython>=7.23.1->ipykernel->jupyter) (4.4.2)

Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-packages (from ipython>=7.23.1->ipykernel->jupyter) (0.7.5)

Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-packages (from ipython>=7.23.1->ipykernel->jupyter) (0.2.0)

Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-packages (from ipython>=7.23.1->ipykernel->jupyter) (4.9.0)

Requirement already satisfied: entrypoints in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->ipykernel->jupyter) (0.4)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->ipykernel->jupyter) (2.9.0.post0)

Requirement already satisfied: platformdirs>=2.5 in /usr/local/lib/python3.12/dist-packages (from jupyter-core!=5.0.*,>=4.1.2->jupyter-console->jupyter) (4.9.4)

Requirement already satisfied: jupyter-events>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (0.12.0)

Requirement already satisfied: jupyter-server-terminals>=0.4.4 in /usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (0.5.4)

Requirement already satisfied: overrides>=5.0 in /usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (7.7.0)

Requirement already satisfied: websocket-client>=1.7 in /usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (1.9.0)

Requirement already satisfied: argon2-cffi-bindings in /usr/local/lib/python3.12/dist-packages (from argon2-cffi->notebook->jupyter) (25.1.0)

Requirement already satisfied: babel>=2.10 in /usr/local/lib/python3.12/dist-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (2.18.0)

Collecting json5>=0.9.0 (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)

 Downloading json5-0.14.0-py3-none-any.whl.metadata (36 kB)

Requirement already satisfied: jsonschema>=4.18.0 in /usr/local/lib/python3.12/dist-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (4.26.0)

Requirement already satisfied: requests>=2.31 in /usr/local/lib/python3.12/dist-packages (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (2.32.4)

Requirement already satisfied: fastjsonschema>=2.15 in /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert->jupyter) (2.21.2)

Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-packages (from prompt-toolkit>=3.0.30->jupyter-console->jupyter) (0.6.0)

Requirement already satisfied: ptyprocess in /usr/local/lib/python3.12/dist-packages (from terminado>=0.8.3->notebook->jupyter) (0.7.0)

Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->nbconvert->jupyter) (2.8.3)

Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->nbconvert->jupyter) (4.15.0)

Requirement already satisfied: parso<0.9.0,>=0.8.4 in /usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython>=7.23.1->ipykernel->jupyter) (0.8.6)

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (26.1.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (2025.9.1)

Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (0.37.0)

Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (0.30.0)

Requirement already satisfied: python-json-logger>=2.0.4 in /usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (4.1.0)

Requirement already satisfied: pyyaml>=5.3 in /usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (6.0.3)

Requirement already satisfied: rfc3339-validator in /usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (0.1.4)

Requirement already satisfied: rfc3986-validator>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (0.1.1)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->jupyter-client>=6.1.12->ipykernel->jupyter) (1.17.0)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.31->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (3.4.6)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.31->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter) (2.5.0)

Requirement already satisfied: cffi>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from argon2-cffi-bindings->argon2-cffi->notebook->jupyter) (2.0.0)

Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi->notebook->jupyter) (3.0)

Requirement already satisfied: fqdn in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (1.5.1)

Requirement already satisfied: isoduration in /usr/local/lib/python3.12/dist-

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packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (20.11.0)
Requirement already satisfied: jsonpointer>1.13 in
/usr/local/lib/python3.12/dist-packages (from jsonschema[format-
nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter) (3.1.1)
Requirement already satisfied: rfc3987-syntax>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from jsonschema[format-
nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter) (1.1.0)
Requirement already satisfied: uri-template in /usr/local/lib/python3.12/dist-
packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (1.3.0)
Requirement already satisfied: webcolors>=24.6.0 in
/usr/local/lib/python3.12/dist-packages (from jsonschema[format-
nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter) (25.10.0)
Requirement already satisfied: lark>=1.2.2 in /usr/local/lib/python3.12/dist-
packages (from rfc3987-syntax>=1.1.0->jsonschema[format-
nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter) (1.3.1)
Requirement already satisfied: arrow>=0.15.0 in /usr/local/lib/python3.12/dist-
packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (1.4.0)
Requirement already satisfied: tzdata in /usr/local/lib/python3.12/dist-packages
(from arrow>=0.15.0->isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter) (2025.3)
Downloading jupyter-1.1.1-py2.py3-none-any.whl (2.7 kB)
Downloading jupyterlab-4.5.6-py3-none-any.whl (12.4 MB)
12.4/12.4 MB
94.9 MB/s eta 0:00:00
Downloading async_lru-2.3.0-py3-none-any.whl (8.4 kB)
Downloading jupyter_lsp-2.3.1-py3-none-any.whl (77 kB)
77.5/77.5 kB
5.3 MB/s eta 0:00:00
Downloading jupyterlab_server-2.28.0-py3-none-any.whl (59 kB)
59.8/59.8 kB
4.3 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
53.8 MB/s eta 0:00:00
Downloading json5-0.14.0-py3-none-any.whl (36 kB)

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[ ]: import os
from google.colab import drive

# Mount Google Drive if it's not already mounted (from previous steps)

```

```

drive.mount('/content/drive', force_remount=True)

# Define the input path for the current notebook and the output path for the
↳PDF file
# Ensure this matches your notebook's name
current_notebook_name = 'Q-learning Tutorial.ipynb'
input_notebook_path = f'/content/drive/MyDrive/Colab Notebooks/
↳{current_notebook_name}'
output_pdf_path = '/content/drive/MyDrive/Q-learning_Tutorial.pdf'

print(f"Input notebook path: {input_notebook_path}")
print(f"Output PDF path: {output_pdf_path}")

```

```

[ ]: # Use nbconvert to convert the notebook to PDF
# The --output option specifies the output file path
# --to pdf will use LaTeX internally, which requires the packages installed
↳above
!jupyter nbconvert "{input_notebook_path}" --to pdf --output "{output_pdf_path}"

print(f"Notebook converted to PDF and saved to: {output_pdf_path}")
print("You can now find 'Q-learning_Tutorial.pdf' in your Google Drive.")

```